Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Sine

$$\sin x = \sin \alpha$$

Cosine

$$\cos x = \cos \alpha$$

Tangent

$$\tan x = \tan \alpha$$

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Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q001 Written

Solve (a) $-2\cos\theta = \sqrt{2}$ and (b) $\tan\theta + \sqrt{3} = 0$, giving the general solution for (b).

Solution

1. Rewrite as $\cos\theta = -\frac{\sqrt{2}}{2}$ and use QII/QIII.
2. For tangent, the reference angle is $\pi/3$ and the period is π.

Final answer

$(a)\ \theta = \frac{3\pi}{4}, \frac{5\pi}{4} \quad (b)\ \theta = \frac{2\pi}{3} + n\pi,\ n \in \mathbb{Z}$

Q002 Written

Solve (a) $2\cos\theta - 1 = 0$ and (b) $\tan\theta - 1 = 0$ for $0 \le \theta \le 2\pi$.

Solution

1. Use the unit circle for cosine and include both turns in the stated interval.
2. For tangent, repeat the $\pi/4$ reference angle after one period π.

Final answer

$(a)\ \theta = \frac{\pi}{3}, \frac{5\pi}{3} \quad (b)\ \theta = \frac{\pi}{4}, \frac{5\pi}{4}$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q003 **Written**

Solve all source equations on $0 \le \theta \le 2\pi$: **(a)** $2\sin\theta+1=0$, **(b)** $2\sin\theta=\sqrt{3}$, **(c)** $3\sin\theta+3=0$, **(d)** $\sqrt{2}\cos\theta+1=0$, **(e)** $2\cos\theta=\sqrt{3}$, **(f)** $2\cos\theta-\sqrt{2}=0$, **(g)** $\tan\theta+1=0$, **(h)** $\sqrt{3}\tan\theta=1$, **(i)** $\sqrt{3}\tan\theta=3$, **and give general solutions for (j)** $2\sin\theta+\sqrt{3}=0$, **(k)** $\cos\theta+1=0$, **(l)** $\sqrt{3}\tan\theta+3=0$.

Solution

1. Put each equation into a single-function form, find the reference angle, then select the quadrants from the sign.
2. Use period 2π for sine/cosine and π for tangent; reject the endpoint 2π when the interval is half-open.

Final answer

$(a)\ \frac{7\pi}{6}, \frac{11\pi}{6}\ (b)\ \frac{\pi}{3}, \frac{2\pi}{3}\ (c)\ \frac{3\pi}{2}\ (d)\ \frac{3\pi}{4}, \frac{5\pi}{4}\ (e)\ \frac{\pi}{6}, \frac{11\pi}{6}\ (f)\ \frac{\pi}{4}, \frac{7\pi}{4}\ (g)\ \frac{3\pi}{4}, \frac{7\pi}{4}\ (h)\ \frac{\pi}{6}, \frac{7\pi}{6}\ (i)\ \frac{\pi}{3}, \frac{4\pi}{3}\ (j)\ \frac{4\pi}{3}+2n\pi, \frac{5\pi}{3}+2n\pi\ (k)\ \pi+2n\pi\ (l)\ \frac{2\pi}{3}+n\pi$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q004 **Written**

Solve $\tan^2\theta - (2+\sqrt{3})\tan\theta + 2\sqrt{3} = 0$ and give the general solution.

Solution

1. Factor the quadratic as $(\tan\theta - 2)(\tan\theta - \sqrt{3}) = 0$.
2. Apply the tangent period π to both roots.

Final answer

$\tan\theta = 2 \text{ or } \tan\theta = \sqrt{3}; \quad \theta = \arctan(2) + n\pi \text{ or } \theta = \frac{\pi}{3} + n\pi$

Q005 **Written**

Solve for $0 \le \theta < 2\pi$: (a) $2\cos\theta > -\sqrt{3}$, (b) $\tan\theta \le \frac{1}{\sqrt{3}}$.

Solution

1. Solve the boundary equation first, then shade the correct arcs on the unit circle.
2. Tangent is undefined at odd multiples of $\pi/2$; those points are never included.

Final answer

$(a)\ 0 \le \theta < \frac{5\pi}{6} \text{ or } \frac{7\pi}{6} < \theta < 2\pi \quad (b)\ 0 \le \theta \le \frac{\pi}{6} \text{ or } \frac{\pi}{2} < \theta \le \frac{7\pi}{6} \text{ or } \frac{3\pi}{2} < \theta < 2\pi$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q006 Written

Solve for $0 \le \theta < 2\pi$: (a) $2\sin\theta \le -1$, (b) $\cos\theta > \frac{1}{2}$, (c) $\tan\theta > 1$.

Solution

1. Mark the equality angles and choose the arcs where the function has the required sign.
2. Open endpoints correspond to strict inequalities and to tangent asymptotes.

Final answer

$(a)\ \frac{7\pi}{6} \le \theta \le \frac{11\pi}{6} \quad (b)\ 0 \le \theta < \frac{\pi}{3} \text{ or } \frac{5\pi}{3} < \theta < 2\pi \quad (c)\ \frac{\pi}{4} < \theta < \frac{\pi}{2} \text{ or } \frac{5\pi}{4} < \theta < \frac{3\pi}{2}$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q007 Written

Solve all source inequalities for $0 \le \theta < 2\pi$: (a) $\sin\theta > \frac{\sqrt{3}}{2}$, (b) $\sin\theta - \frac{\sqrt{2}}{2} > 0$, (c) $2\sin\theta + \sqrt{3} \le 0$, (d) $\cos\theta > 0$, (e) $\cos\theta - \frac{1}{2} \le 0$, (f) $2\cos\theta + \sqrt{2} \le 0$, (g) $\tan\theta \le -\frac{1}{\sqrt{3}}$, (h) $\tan\theta + \sqrt{3} > 0$, (i) $\tan\theta + 1 \ge 0$.

Solution

1. For each item, solve the equality, draw the relevant horizontal line, and select the arcs satisfying the sign.
2. Keep the distinction between open and closed endpoints and remove all tangent asymptotes.

Final answer

$(a)\ (\frac{\pi}{3}, \frac{2\pi}{3})\ (b)\ (\frac{\pi}{4}, \frac{3\pi}{4})\ (c)\ [\frac{4\pi}{3}, \frac{5\pi}{3}]\ (d)\ [0, \frac{\pi}{2}) \cup (\frac{3\pi}{2}, 2\pi)\ (e)\ [\frac{\pi}{3}, \frac{5\pi}{3}]\ (f)\ [\frac{3\pi}{4}, \frac{5\pi}{4}]\ (g)\ (\frac{\pi}{2}, \frac{5\pi}{6}] \cup (\frac{3\pi}{2}, \frac{11\pi}{6}]\ (h)\ [0, \frac{\pi}{2}) \cup (\frac{2\pi}{3}, \frac{3\pi}{2}) \cup (\frac{5\pi}{3}, 2\pi)\ (i)\ [0, \frac{\pi}{2}) \cup [\frac{3\pi}{4}, \frac{3\pi}{2}) \cup [\frac{7\pi}{4}, 2\pi)$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q008 **Written**

Solve $2\sin^2\theta - \cos\theta - 1 > 0$ **for** $0 \le \theta < 2\pi$**.**

Solution

1. Use $\sin^2\theta = 1 - \cos^2\theta$ to obtain $(1 - 2\cos\theta)(1 + \cos\theta) > 0$.
2. Since $-1 \le \cos\theta \le 1$, the solution is $-1 < \cos\theta < 1/2$; exclude $\theta = \pi$ where the product is zero.

Final answer

$\frac{\pi}{3} < \theta < \pi$ or $\pi < \theta < \frac{5\pi}{3}$

Q009 **Written**

Solve for $0 \le \theta < 2\pi$**: (a)** $2\cos^2\theta - 3\sin\theta = 0$**, (b)** $2\sin^2\theta - 3\cos\theta - 3 \le 0$**.**

Solution

1. For (a), replace $\cos^2\theta$ and solve $2s^2 + 3s - 2 = 0$; reject $s = -2$.
2. For (b), replace $\sin^2\theta$, solve $(2c + 1)(c + 1) \ge 0$, and intersect with $[-1, 1]$.

Final answer

$(a)\ \theta = \frac{\pi}{6}, \frac{5\pi}{6} \quad (b)\ 0 \le \theta \le \frac{2\pi}{3}$ or $\theta = \pi$ or $\frac{4\pi}{3} \le \theta < 2\pi$

Worked Solutions Solving Equations and Inequalities Involving Trigonometric Functions

Q010 Written

Solve for $0 \le \theta < 2\pi$: (a) $2\sin^2\theta - \sin\theta = 0$, (b) $2\cos^2\theta - \sin\theta = 2$, (c) $2\sin^2\theta - 4 < 5\cos\theta$.

Solution

1. Set $s = \sin\theta$ or $c = \cos\theta$ after using $s^2 + c^2 = 1$.
2. Solve the resulting quadratic, reject impossible trig values, then read the unit-circle intervals.

Final answer

$(a)\ \theta = 0, \dfrac{\pi}{6}, \dfrac{5\pi}{6}, \pi \quad (b)\ \theta = 0, \pi, \dfrac{7\pi}{6}, \dfrac{11\pi}{6} \quad (c)\ 0 \le \theta < \dfrac{2\pi}{3} \text{ or } \dfrac{4\pi}{3} < \theta < 2\pi$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q011 **Written**

Solve all source equations and inequalities for $0 \le \theta < 2\pi$: (a) $2\sin^2\theta + \sin\theta - 1 = 0$, (b) $2\sin^2\theta + 3\cos\theta = 0$, (c) $2\sin^2\theta - \cos\theta = 2$, (d) $2\cos^2\theta - 5\cos\theta - 3 = 0$, (e) $2\cos^2\theta + \sin\theta - 1 = 0$, (f) $2\cos^2\theta + 3\sin\theta = 3$, (g) $2\cos^2\theta + 3\sin\theta \le 0$, (h) $2\sin^2\theta + \cos\theta > 1$, (i) $2\cos^2\theta - 4 \le 5\sin\theta$.

Solution

1. Convert to one trig function, solve the quadratic, and apply $-1 \le \sin\theta, \cos\theta \le 1$.
2. For inequalities, use the sign chart in the trig variable and then translate the accepted interval(s) to θ.

Final answer

$(a)\ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}\ (b)\ \frac{2\pi}{3}, \frac{4\pi}{3}\ (c)\ \frac{2\pi}{3}, \frac{\pi}{2}, \frac{4\pi}{3}, \frac{3\pi}{2}\ (d)\ \frac{2\pi}{3}, \frac{4\pi}{3}\ (e)\ \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}\ (f)\ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}\ (g)\ \frac{7\pi}{6} \le \theta \le \frac{11\pi}{6}\ (h)\ 0 < \theta < \frac{2\pi}{3} \text{ or } \frac{4\pi}{3} < \theta < 2\pi\ (i)\ 0 \le \theta \le \frac{7\pi}{6} \text{ or } \frac{11\pi}{6} \le \theta < 2\pi$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q012 Written

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ for $0 \leq \theta < 2\pi$.

Solution

1. Subtract 1 and simplify the numerator to $\sin\theta(1-\sin\theta)$.
2. The denominator is positive but undefined at $\theta = \pi/2$; hence $0 \leq \sin\theta \leq 1$ with that point removed.

Final answer

$0 \leq \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \leq \pi$

Q013 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, (b) $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

Solution

1. Let $t = \theta + \alpha$, write the range of t, solve there, then subtract α.
2. For the inequality, shade one complete revolution in the shifted t-range and preserve strict endpoints.

Final answer

$(a)\ \theta = \frac{\pi}{2}, \frac{11\pi}{6} \quad (b)\ 0 \leq \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$

Worked Solutions | **Solving Equations and Inequalities Involving Trigonometric Functions**

Q014 **Written**

Solve for $0 \le \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, (b) $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, (c) $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$.

Solution

1. Find each shifted argument's exact range before selecting the unit-circle arcs.
2. For tangent, exclude asymptotes automatically; for strict cosine inequality, endpoints are open.

Final answer

$(a)\ \theta = 0, \frac{2\pi}{3} \quad (b)\ \theta = \pi 12, \frac{13\pi}{12} \quad (c)\ \frac{\pi}{2} < \theta < \pi$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q015 **Written**

Solve for $0 \le \theta < 2\pi$**: (a)** $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$**, (b)** $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$**, (c)** $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$**, (d)** $\tan(\theta + \frac{\pi}{4}) = -1$**, (e)** $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$**, (f)** $\tan(\theta + \frac{\pi}{6}) = 1$**, (g)** $\sin(\theta + \frac{\pi}{6}) \le \frac{\sqrt{3}}{2}$**, (h)** $\sin(\theta + \frac{\pi}{4}) \ge \frac{\sqrt{2}}{2}$**, (i)** $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$**.**

Solution

1. Set $t = \theta + \alpha$, record its interval, solve the standard trig relation in that interval, then subtract α.
2. Check every endpoint against the original strict/non-strict sign and the domain $0 \le \theta < 2\pi$.

Final answer

$(a)\ \frac{\pi}{2}, \frac{5\pi}{6}\ (b)\ \frac{\pi}{2}, \frac{7\pi}{6}\ (c)\ \pi 12, \frac{5\pi}{12}\ (d)\ \frac{\pi}{2}, \frac{3\pi}{2}\ (e)\ \frac{5\pi}{6}, \frac{11\pi}{6}\ (f)\ \pi 12, \frac{13\pi}{12}\ (g)\ [0, \frac{\pi}{6}] \cup [\frac{\pi}{2}, 2\pi)\ (h)\ [0, \frac{\pi}{2}]\ (i)\ \frac{4\pi}{3} < \theta < 2\pi$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q016 **Written**

Solve $4\sin^2(\theta + \frac{\pi}{6}) \geq 1$ **for** $0 \leq \theta < 2\pi$.

Solution

1. Let $t = \theta + \pi/6$; then $|\sin(t)| \geq 1/2$.
2. On the shifted range $[\pi/6, 13\pi/6)$, the accepted arcs give the stated closed θ-intervals.

Final answer

$0 \leq \theta \leq \frac{2\pi}{3}$ or $\pi \leq \theta \leq \frac{5\pi}{3}$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q017 Written

A student divides $\tan x \times \cos x = \frac{1}{2}$ by $\cos x$, obtaining $\tan x = \frac{1}{2\cos x}$. Evaluate this strategy and propose a better substitution on $[0°, 360°[$.

Solution

1. Replace $\tan(x)$ by $\sin(x)/\cos(x)$ before cancelling; the original tangent already excludes $\cos(x) = 0$.
2. Solve the resulting sine equation on the stated degree interval.

Final answer

Dividing by $\cos(x)$ separates the interacting factors and is not helpful; use $\tan(x)\cos(x) = \sin(x)$ (with $\cos(x) \neq 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30°, 150°$.

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q018 MCQ

The period used for a general tangent solution is:

Solution

1. Tangent repeats after π.

Correct option: B

Final answer

π

Q019 MCQ

The solutions of $2\cos\theta = 1$ on $[0, 2\pi]$ are:

Solution

1. Cosine is $1/2$ in QI and QIV.

Correct option: B

Final answer

$\pi/3, 5\pi/3$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q020 *MCQ*

Which root is impossible when solving $2\sin^2\theta + 3\sin\theta - 2 = 0$?

Solution

1. A sine value must lie in $[-1, 1]$.

Correct option: B

Final answer

$\sin\theta = -2$

Q021 *MCQ*

The quadratic challenge factors into roots:

Solution

1. Use $(u-2)(u-\sqrt{3})$ with $u = \tan\theta$.

Correct option: B

Final answer

$2, \sqrt{3}$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q022 MCQ

For $2\cos\theta > -\sqrt{3}$, the excluded arc is:

Solution

1. Cosine is at most $-\sqrt{3}/2$ on the central QII arc.

Correct option: B

Final answer

$[5\pi/6, 7\pi/6]$

Q023 MCQ

The inequality $\tan\theta > 1$ on $[0, 2\pi)$ holds on:

Solution

1. Use the increasing tangent branches and omit asymptotes.

Correct option: A

Final answer

$(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q024 *MCQ*

A closed endpoint in a trig inequality is used when the sign is:

Solution

1. Equality is included for $\leq$ or $\geq$.

Correct option: C

Final answer

greater than or equal to

Q025 *MCQ*

The 5-14 challenge reduces to which cosine condition?

Solution

1. Factor the expression after replacing $\sin^2\theta$.

Correct option: B

Final answer

$-1 < \cos\theta < 1/2$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q026 MCQ

When a quadratic gives $\sin\theta = -2$, we:

Solution

1. Sine cannot be outside $[-1, 1]$.

Correct option: B

Final answer

reject it

Q027 MCQ

The first step for $2\cos^2\theta - 3\sin\theta = 0$ is to use:

Solution

1. Rewrite in one trig function.

Correct option: A

Final answer

$\sin^2\theta + \cos^2\theta = 1$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q028 MCQ

The solution of $2\sin^2\theta + \sin\theta - 1 = 0$ includes:

Solution

1. The valid root $\sin\theta = 1/2$ gives $\pi/6, 5\pi/6$.

Correct option: A

Final answer

$\pi/6$

Q029 MCQ

In the rational challenge, which angle must be excluded?

Solution

1. The denominator $\cos^2\theta$ is zero there.

Correct option: C

Final answer

$\pi/2$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q030 MCQ

After setting $t = \theta + \pi/3$ with $0 \leq \theta < 2\pi$, the t-range is:

Solution

1. Add $\pi/3$ to both domain endpoints.

Correct option: B

Final answer

$[\pi/3, 7\pi/3)$

Q031 MCQ

The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ include:

Solution

1. Set the shifted argument to $\pi/3$.

Correct option: A

Final answer

$\pi/12$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q032 MCQ

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ-solution is:

Solution

1. The shifted sine is at least $\sqrt{2}/2$ from $\pi/4$ to $3\pi/4$.

Correct option: A

Final answer

$[0, \pi/2]$

Q033 MCQ

The 5-16 challenge is equivalent to:

Solution

1. Divide by 4 and take the square-root inequality.

Correct option: A

Final answer

$|\sin t| \geq 1/2$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Q034 **MCQ**

In the Reflect problem, $\tan x \cos x$ **simplifies to:**

Solution

1. Use $\tan(x) = \sin(x)/\cos(x)$ where defined.

Correct option: B

Final answer

$\sin x$

Quiz MCQ 1 **Concept Quiz · MCQ**

The general solution of $\tan\theta = -1$ **is:**

Solution

1. Tangent has period π.

Correct option: B

Final answer

$\theta = -\frac{\pi}{4} + n\pi$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Quiz MCQ 2 **Concept Quiz · MCQ**

The solution of $\cos\theta > 0$ on $[0, 2\pi)$ is:

Solution

1. Cosine is positive in QI and QIV.

Correct option: B

Final answer

$[0, \pi/2) \cup (3\pi/2, 2\pi)$

Quiz MCQ 3 **Concept Quiz · MCQ**

The valid sine root of $2\sin^2\theta + 3\sin\theta - 2 = 0$ is:

Solution

1. The other algebraic root is outside the sine range.

Correct option: B

Final answer

$\sin\theta = \dfrac{1}{2}$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Quiz MCQ 4 **Concept Quiz · MCQ**

The first operation in a shifted-argument problem is to:

Solution

1. The t-range prevents losing or adding solutions.

Correct option: B

Final answer

set t equal to the whole argument and find its range

Quiz WRITTEN 5 **Concept Quiz · Written**

Solve $2\cos\theta = \sqrt{3}$ for $0 \le \theta < 2\pi$.

Solution

1. Cosine is $\sqrt{3}/2$ in QI and QIV.

Final answer

$\theta = \dfrac{\pi}{6}, \dfrac{11\pi}{6}$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Quiz WRITTEN 6 Concept Quiz · Written

Solve $\tan\theta \le -1$ **for** $0 \le \theta < 2\pi$**.**

Solution

1. Use the two tangent branches and include the equality angles.

Final answer

$\frac{\pi}{2} < \theta \le \frac{3\pi}{4}$ or $\frac{3\pi}{2} < \theta \le \frac{7\pi}{4}$

Quiz WRITTEN 7 Concept Quiz · Written

Solve $2\sin^2\theta + \sin\theta - 1 = 0$ **for** $0 \le \theta < 2\pi$**.**

Solution

1. Factor to $(2\sin\theta - 1)(\sin\theta + 1) = 0$, then use the unit circle.

Final answer

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

Worked Solutions **Solving Equations and Inequalities Involving Trigonometric Functions**

Quiz WRITTEN 8 **Concept Quiz · Written**

Solve $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ **for** $0 \leq \theta < 2\pi$.

Solution

1. Set $t = \theta + \pi/3$, solve $\cos(t) > 1/2$ in $[\pi/3, 7\pi/3)$, then subtract.

Final answer

$\frac{4\pi}{3} < \theta < 2\pi$

Answer key

Use the question number shown on each page.

Q001 $(a)\ \theta = \frac{3\pi}{4}, \frac{5\pi}{4}$ $(b)\ \theta = \frac{2\pi}{3} + n\pi,\ n \in \mathbb{Z}$

Q002 $(a)\ \theta = \frac{\pi}{3}, \frac{5\pi}{3}$ $(b)\ \theta = \frac{\pi}{4}, \frac{5\pi}{4}$

Q003 $(a)\ \frac{7\pi}{6}, \frac{11\pi}{6}$ $(b)\ \frac{\pi}{3}, \frac{2\pi}{3}$ $(c)\ \frac{3\pi}{2}$ $(d)\ \frac{3\pi}{4}, \frac{5\pi}{4}$ $(e)\ \frac{\pi}{6}, \frac{11\pi}{6}$ $(f)\ \frac{\pi}{4}, \frac{7\pi}{4}$ $(g)\ \frac{3\pi}{4}, \frac{7\pi}{4}$ $(h)\ \frac{\pi}{6}, \frac{7\pi}{6}$ $(i)\ \frac{\pi}{3}, \frac{4\pi}{3}$ $(j)\ \frac{4\pi}{3} + 2n\pi, \frac{5\pi}{3} + 2n\pi$ $(k)\ \pi + 2n\pi$ $(l)\ \frac{2\pi}{3}$ [illegible]

Q004 $\tan\theta = 2$ or $\tan\theta = \sqrt{3}$; $\theta = \arctan(2) + n\pi$ or $\theta = \frac{\pi}{3} + n\pi$

Q005 $(a)\ 0 \le \theta < \frac{5\pi}{6}$ or $\frac{7\pi}{6} < \theta < 2\pi$ $(b)\ 0 \le \theta \le \frac{\pi}{6}$ or $\frac{\pi}{2} < \theta \le \frac{7\pi}{6}$ or $\frac{3\pi}{2} < \theta < 2\pi$

Q006 $(a)\ \frac{7\pi}{6} \le \theta \le \frac{11\pi}{6}$ $(b)\ 0 \le \theta < \frac{\pi}{3}$ or $\frac{5\pi}{3} < \theta < 2\pi$ $(c)\ \frac{\pi}{4} < \theta < \frac{\pi}{2}$ or $\frac{5\pi}{4} < \theta < \frac{3\pi}{2}$

Q007 $(a)\ (\frac{\pi}{3}, \frac{2\pi}{3})$ $(b)\ (\frac{\pi}{4}, \frac{3\pi}{4})$ $(c)\ [\frac{4\pi}{3}, \frac{5\pi}{3}]$ $(d)\ [0, \frac{\pi}{2}) \cup (\frac{3\pi}{2}, 2\pi)$ $(e)\ [\frac{\pi}{3}, \frac{5\pi}{3}]$ $(f)\ [\frac{3\pi}{4}, \frac{5\pi}{4}]$ $(g)\ (\frac{\pi}{2}, \frac{5\pi}{6}] \cup (\frac{3\pi}{2}, \frac{11\pi}{6}]$ $(h)\ [0, \frac{\pi}{2}) \cup (\frac{2\pi}{3}, \frac{3\pi}{2}) \cup (\frac{5\pi}{3}, 2\pi)$ $(i)\ [0, \frac{\pi}{2}) \cup [\frac{3\pi}{4}, \frac{3\pi}{2}) \cup [\frac{7\pi}{4}, 2\pi)$

Q008 $\frac{\pi}{3} < \theta < \pi$ or $\pi < \theta < \frac{5\pi}{3}$

Q009 $(a)\ \theta = \frac{\pi}{6}, \frac{5\pi}{6}$ $(b)\ 0 \le \theta \le \frac{2\pi}{3}$ or $\theta = \pi$ or $\frac{4\pi}{3} \le \theta < 2\pi$

Q010 $(a)\ \theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$ $(b)\ \theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$ $(c)\ 0 \le \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$

Q011 $(a)\ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ $(b)\ \frac{2\pi}{3}, \frac{4\pi}{3}$ $(c)\ \frac{2\pi}{3}, \frac{\pi}{2}, \frac{4\pi}{3}, \frac{3\pi}{2}$ $(d)\ \frac{2\pi}{3}, \frac{4\pi}{3}$ $(e)\ \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ $(f)\ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$ $(g)\ \frac{7\pi}{6} \le \theta \le \frac{11\pi}{6}$ $(h)\ 0 < \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ $(i)\ 0 \le \theta \le \frac{7\pi}{6}$ or $\frac{11\pi}{6} \le \theta < 2\pi$

Q012 $0 \le \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \le \pi$

Q013 $(a)\ \theta = \frac{\pi}{2}, \frac{11\pi}{6}$ $(b)\ 0 \le \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$

Q014 $(a)\ \theta = 0, \frac{2\pi}{3}$ $(b)\ \theta = \pi 12, \frac{13\pi}{12}$ $(c)\ \frac{\pi}{2} < \theta < \pi$

Q015 $(a)\ \frac{\pi}{2}, \frac{5\pi}{6}$ $(b)\ \frac{\pi}{2}, \frac{7\pi}{6}$ $(c)\ \pi 12, \frac{5\pi}{12}$ $(d)\ \frac{\pi}{2}, \frac{3\pi}{2}$ $(e)\ \frac{5\pi}{6}, \frac{11\pi}{6}$ $(f)\ \pi 12, \frac{13\pi}{12}$ $(g)\ [0, \frac{\pi}{6}] \cup [\frac{\pi}{2}, 2\pi)$ $(h)\ [0, \frac{\pi}{2}]$ $(i)\ \frac{4\pi}{3} < \theta < 2\pi$

Q016 $0 \le \theta \le \frac{2\pi}{3}$ or $\pi \le \theta \le \frac{5\pi}{3}$

Q017 Dividing by $\cos(x)$ separates the interacting factors and is not helpful; use $\tan(x)\cos(x) = \sin(x)$ (with $\cos(x) \ne 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30°, 150°$.

Q018 B

Q019 B

Q020 B

Q021 B

Q022 B

Q023 A

Q024 C

Q025 B

Q026 B

Q027 A

Q028 A

Q029 C

Q030 B

Q031 A

Q032 A

Q033 A

Q034 B

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $\theta = \frac{\pi}{6}, \frac{11\pi}{6}$

Quiz WRITTEN 6 $\frac{\pi}{2} < \theta \le \frac{3\pi}{4}$ or $\frac{3\pi}{2} < \theta \le \frac{7\pi}{4}$

Quiz WRITTEN 7 $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

Quiz WRITTEN 8 $\frac{4\pi}{3} < \theta < 2\pi$